

Interest Rates and Yield Curves

Corporate Finance Chapter 6

I. Determinants of Interest Rates

The required return on a debt security is determined by several components:

Interest Rate Equation

$$r = r^* + IP + DRP + LP + MRP$$

	Component	Description
Where:	r	Required return on a debt security
	r^*	Real risk-free rate of interest
	IP	Inflation premium
	DRP	Default risk premium
	LP	Liquidity premium
	MRP	Maturity risk premium

Premiums Added to r^* for Different Types of Debt

Debt Type	IP	MRP	DRP	LP
Short-Term Treasury	✓			
Long-Term Treasury	✓	✓		
Short-Term Corporate	✓		✓	✓
Long-Term Corporate	✓	✓	✓	✓

II. Yield Curve and Term Structure

- **Term Structure:** The relationship between interest rates (or yields) and maturities.
- **Yield Curve:** A graph of the term structure.

Constructing the Yield Curve

Step 1: Calculate Inflation Premium (IP)

The inflation premium for a security with N years to maturity is the average expected inflation rate over Years 1 to N:

$$IP_N = \frac{\sum_{t=1}^N INFL_t}{N}$$

Example: Inflation Expectations

Assume inflation is expected to be:

- Year 1: 5%
- Year 2: 6%
- Year 3 onward: 8%

$$IP_1 = 5\%/1 = 5.00\%$$

$$IP_{10} = [5\% + 6\% + 8\% \times 8]/10 = 75\%/10 = 7.50\%$$

$$IP_{20} = [5\% + 6\% + 8\% \times 18]/20 = 155\%/20 = 7.75\%$$

Step 2: Calculate Maturity Risk Premium (MRP)

For this example, the maturity risk premium is calculated as:

$$MRP_t = 0.1\% \times (t - 1)$$

$$MRP_1 = 0.1\% \times (1 - 1) = 0.0\%$$

$$MRP_{10} = 0.1\% \times (10 - 1) = 0.9\%$$

$$MRP_{20} = 0.1\% \times (20 - 1) = 1.9\%$$

Step 3: Calculate Nominal Rates

Assuming $r^* = 3\%$:

$$r_{RF,t} = r^* + IP_t + MRP_t$$

$$r_{RF,1} = 3\% + 5.00\% + 0.0\% = 8.0\%$$

$$r_{RF,10} = 3\% + 7.50\% + 0.9\% = 11.4\%$$

$$r_{RF,20} = 3\% + 7.75\% + 1.9\% = 12.65\%$$

Hypothetical Yield Curve

The resulting yield curve is **upward sloping** due to:

- Increasing expected inflation
- Increasing maturity risk premium

III. Corporate vs. Treasury Yield Curves

Key Relationships

- Corporate yield curves are **higher** than Treasury yield curves.
- The spread between corporate and Treasury curves **widens** as the corporate bond rating decreases.
- Corporate yields include **DRP** (default risk premium) and **LP** (liquidity premium).

$$\text{Corporate Bond Yield Spread} = \text{Corporate Bond Yield} - \text{Treasury Bond Yield} = \text{DRP} + \text{LP}$$

Representative Interest Rates on 5-Year Bonds (May 2011)

Issuer	Rate	Spread = DRP + LP
U.S. Treasury	1.81%	—
AAA Corporate	2.00%	0.19%
AA Corporate	2.68%	0.87%
A Corporate	2.79%	0.98%

Yield Curve Comparison: Treasury vs. Corporate

Maturity	Treasury Bond	AA-Rated Bond	BBB-Rated Bond
1 year	5.5%	6.7%	7.4%
5 years	6.1%	7.4%	8.1%
10 years	6.8%	8.2%	9.1%
20 years	7.4%	9.2%	10.2%
30 years	7.7%	9.8%	11.1%

IV. Pure Expectations Theory

Theory Overview

- The shape of the yield curve depends on **investors' expectations about future interest rates**.
- If interest rates are expected to increase, long-term rates will be higher than short-term rates (upward sloping).
- If interest rates are expected to decrease, long-term rates will be lower than short-term rates (downward sloping).

Key Assumptions

1. The maturity risk premium for Treasury securities is **zero** ($MRP = 0$).
2. Long-term rates are an **average of current and future short-term rates**.
3. The yield curve can be used to "back out" expected future interest rates.

Example: Observed Treasury Rates

Maturity	Yield
1 year	6.0%
2 years	6.2%
3 years	6.4%
4 years	6.5%
5 years	6.5%

Calculating Forward Rates

One-Year Forward Rate (One Year from Now)

To find the expected interest rate on a one-year security, one year from now:

$$(1.062)^2 = (1.060)(1 + X)$$

$$\frac{1.12784}{1.060} = 1 + X$$

$$X = 6.4004\%$$

Alternative (Arithmetic Average):

$$6.2\% = \frac{6.0\% + X}{2} \Rightarrow X = 6.4\%$$

Three-Year Forward Rate (Two Years from Now)

To find the expected interest rate on a three-year security, two years from now:

$$(1.065)^5 = (1.062)^2(1 + X)^3$$

$$\frac{1.37009}{1.12784} = (1 + X)^3$$

$$X = 6.7005\%$$

Criticisms of Pure Expectations Theory

- Most evidence suggests that $MRP \neq 0$.
- Lenders generally prefer short-term securities and view long-term securities as riskier.

- Investors demand a premium to hold long-term securities ($MRP > 0$).

V. Macroeconomic Factors Influencing Interest Rate Levels

1. **Federal Reserve Policy:** Central bank actions affect money supply and short-term rates.
2. **Federal Budget Deficits or Surpluses:** Government borrowing affects demand for funds.
3. **International Factors:** Global capital flows, foreign investment, and exchange rates.
4. **Level of Business Activity:** Economic growth affects demand for credit.

Summary of Key Formulas

$$r = r^* + IP + DRP + LP + MRP$$

$$IP_N = \frac{\sum_{t=1}^N INFL_t}{N}$$

$$MRP_t = 0.1\% \times (t - 1) \quad (\text{example formula})$$

$$r_{RF,t} = r^* + IP_t + MRP_t$$

$$\text{Corporate Spread} = DRP + LP$$

$$\text{Forward Rate: } (1 + r_N)^N = (1 + r_{N-1})^{N-1}(1 + f_{N-1,1})$$